$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ and } D = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

By solving we get

$$\boxed{X=(I-B)^{-1}D}$$

The matrix B is known as the ***Technology matrix***.

1.3.1 The Hawkins – Simon conditions

Hawkins – Simon conditions ensure the viability of the system.

If B is the technology matrix then Hawkins – Simon conditions are

(i) the main diagonal elements in $I - B$ must be positive and

(ii) $|I - B|$ must be positive.

Example 1.23

The technology matrix of an economic system of two industries is $\begin{bmatrix} 0.8 & 0.2 \\ 0.9 & 0.7 \end{bmatrix}$. Test whether the system is viable as per Hawkins – Simon conditions.

Solution

$$\text{B} = \begin{bmatrix} 0.8 & 0.2 \\ 0.9 & 0.7 \end{bmatrix}$$

$$I - B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.8 & 0.2 \\ 0.9 & 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.2 & -0.2 \\ -0.9 & 0.3 \end{bmatrix}$$

$$|I - B| = \begin{vmatrix} 0.2 & -0.2 \\ -0.9 & 0.3 \end{vmatrix}$$

$$= (0.2)(0.3)-(-0.2)(-0.9)$$

$$= 0.06 - 0.18$$

$$= -0.12 < 0$$

Since $|I - B|$ is negative, Hawkins – Simon conditions are not satisfied.

Therefore, the given system is not viable.

Example 1.24

The following inter – industry transactions table was constructed for an economy of the year 2016.

Industry	1	2	Final consumption	Total output
1	500	1,600	400	2,500
2	1,750	1,600	4,650	8,000
Labours	250	4,800	-	-

Construct technology co-efficient matrix showing direct requirements. Does a solution exist for this system.

Solution

$$a_{11} = 500 \quad a_{12} = 1600 \quad x_1 = 2500$$

$$a_{21} = 1750 \quad a_{22} = 1600 \quad x_2 = 8000$$

$$b_{11} = \frac{a_{11}}{x_1} = \frac{500}{2500} = 0.20;$$

$$b_{12} = \frac{a_{12}}{x_2} = \frac{1600}{8000} = 0.20$$

$$b_{21} = \frac{a_{21}}{x_1} = \frac{1750}{2500} = 0.70;$$

$$b_{22} = \frac{a_{22}}{x_2} = \frac{1600}{8000} = 0.20$$

The technology matrix is $= \begin{bmatrix} 0.2 & 0.2 \\ 0.7 & 0.2 \end{bmatrix}$

$$I - B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.2 & 0.2 \\ 0.7 & 0.2 \end{bmatrix}$$

$= \begin{bmatrix} 0.8 & -0.2 \\ -0.7 & 0.8 \end{bmatrix}$, elements of main diagonal are positive

$$\text{Now, } |I - B| = \begin{vmatrix} 0.8 & -0.2 \\ -0.7 & 0.8 \end{vmatrix}$$

$$= (0.8)(0.8)-(-0.7)(-0.2)$$

$$= 0.64 - 0.14$$

$$= 0.50 > 0$$

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